

Master Research Reference v2

Corrected source of truth for execution, controls, and Duck checkpoints

Cross-Cycle Explanation Drift in Solar Flare Forecasting

Version 2.0 | October 2026 | GitHub Base Research Documentation

Status	REVIEWED / BASELINE LOCKED
Research domain	Space Weather Machine Learning Explainable AI

1. Executive Research Lock

FINAL RESEARCH STATEMENT

Train solar-flare prediction models on Cycle-24-era definitive SHARP data, freeze the full prediction pipeline, then evaluate the same models on an independent Cycle-24 holdout and pre-defined Cycle-25 temporal blocks. Compare predictive skill and SHAP attribution distributions. Treat cross-cycle explanation drift as evidence only when it exceeds within-Cycle-24 sampling and temporal controls and persists under pre-specified shift-aware and correlated-feature sensitivity analyses. SHAP describes model attribution, not causal solar physics.

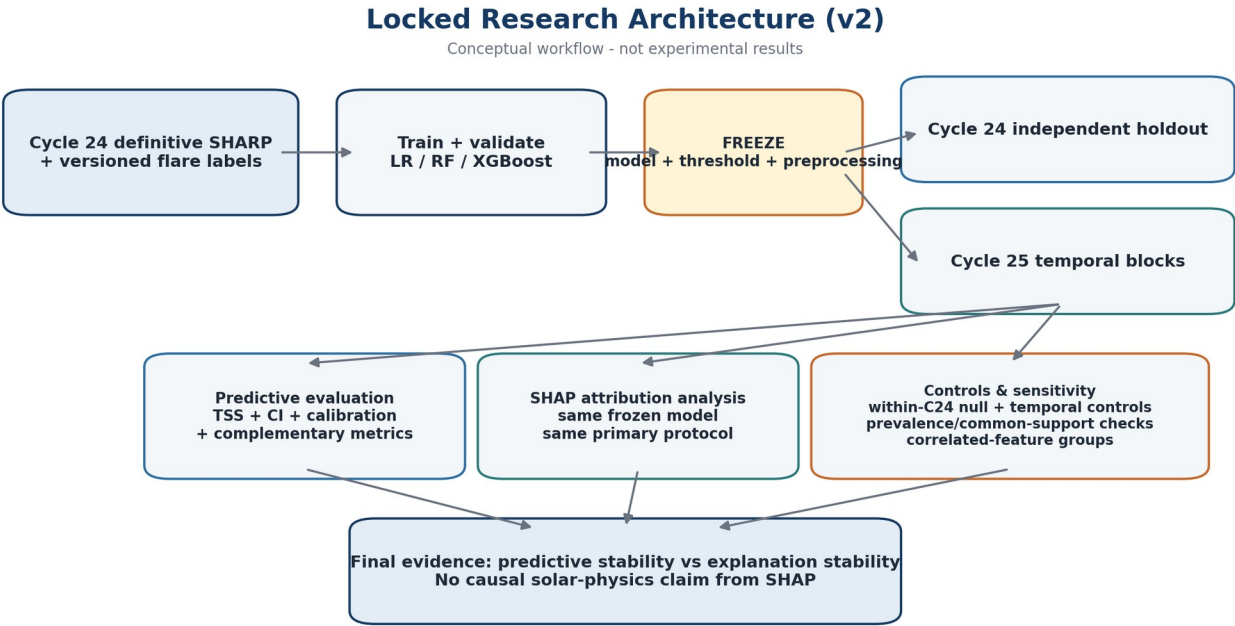


Figure 1. Corrected cross-cycle research architecture. No experimental result is implied.

2. Corrections Applied to Master v1

v1 issue	v2 correction
Cycle 25 called OOD	Use "cross-cycle temporal/domain-shift evaluation" unless support tests justify stronger OOD language.
HARP treated as equivalent to NOAA AR	Use connected HARP-NOAA entity groups because a HARP can map to zero, one, or multiple NOAA ARs.
Generic SHARP source	Primary product: definitive SHARP; lock exact JSOC series (preferably hmi.sharp_cea_720s) and QUALITY rules before extraction.
Cadence under-specified	Record native 12-min cadence and pre-register model sampling stride / snapshot cadence before outcome analysis.

v1 issue	v2 correction
Distribution shift only audited	Add class-conditional, prevalence-standardized, and common-support / reweighting sensitivity analyses.
Null baseline only resampling	Add same-cycle temporal controls in Cycle 24, not just bootstrap resampling.
"Stable" / "drift" not numerically defined	Pre-register equivalence / change rules and empirical null statistics before opening Cycle-25 results.
Model said to behave "internally differently"	A frozen model does not change parameters; the attribution distribution over new inputs may change.
Label source sensitivity absent	Use one versioned primary flare catalog and, if feasible, a secondary label-source sensitivity analysis.

3. Locked Scientific Question

Primary question: When the same frozen Cycle-24-trained flare-prediction model is evaluated on Cycle 25, do predictive skill and feature-attribution distributions remain stable, or can they drift independently?

Secondary question: If attribution changes are observed, do they exceed ordinary within-Cycle-24 sampling and temporal variability, and do they persist under controls for prevalence, covariate support, label-source sensitivity, and correlated SHARP features?

Terminology rule

"Explanation drift" means a statistically supported change in the distribution/structure of SHAP attributions under a fixed model and fixed explanation protocol. It does NOT mean the model parameters changed, and it does NOT prove changed solar physics.

4. Locked Data Design

Element	Locked / required rule
Magnetic data	SDO/HMI definitive SHARP; exact JSOC series and access metadata versioned. Native SHARP cadence is 12 minutes.
Preferred series	hmi.sharp_cea_720s for a consistent definitive CEA-based research product; final series locked before extraction.
Training window	Cycle-24-era HMI/SHARP window: May 2010 through December 2019.
Transfer window	Cycle 25: 2020 onward, divided into pre-defined temporal blocks.
Target	Binary forecast: $\geq$ M-class flare in the next 24 h, after the event-association rule is locked.

Element	Locked / required rule
Sampling stride	Pre-register before final extraction/model fitting. Native records are 12-min; a coarser research stride may be used to reduce redundancy.
Labels	One primary versioned GOES/NOAA flare catalog; optional secondary-source sensitivity analysis.
Quality	Definitive product, QUALITY filtering, disk-position/limb rule, missingness handling, and query provenance saved.

4.1 Leakage-safe entity definition

A HARP is an automatically tracked magnetic-flux complex and is not in one-to-one correspondence with NOAA active-region numbers. Therefore, the split unit is an entity group built from connected HARP and NOAA identifiers. All linked identifiers are assigned to exactly one train/validation/holdout partition. HARPs with no NOAA association remain standalone groups.

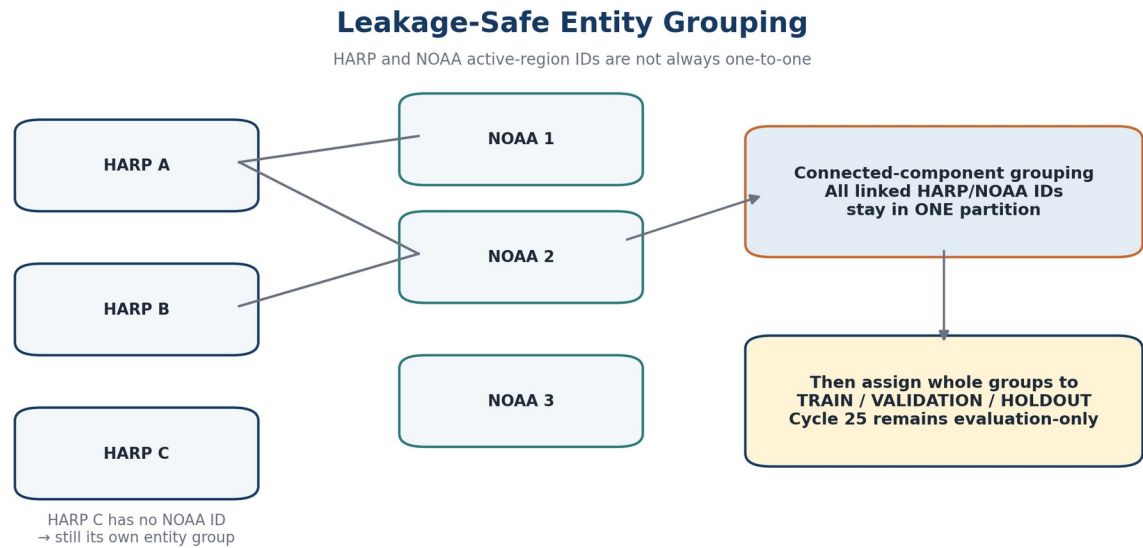
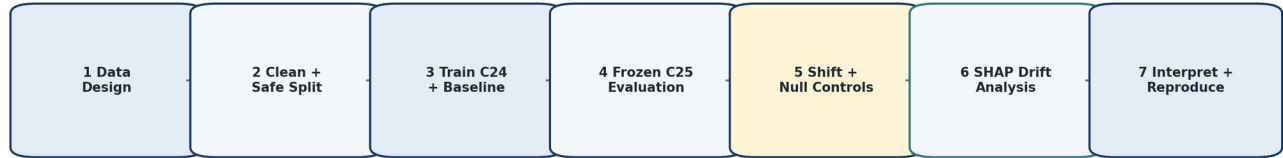


Figure 2. Entity-group split logic preventing HARP/NOAA cross-link leakage.

5. Seven-Phase Execution Plan

Seven-Phase Execution Pipeline



Phase gates: each phase has required outputs + an exit criterion before moving forward

Figure 3. Seven-phase research execution plan.

Phase	Name	Required work
1	Data collection & design	Lock data product, label source, cadence/stride, forecast target, association rules, provenance.
2	Cleaning & leakage-safe split	Build connected HARP-NOAA entity groups; fit preprocessing only on Cycle-24 training data.
3	Cycle 24 training & baseline	Train LR/RF/XGBoost; tune only on Cycle-24 validation; lock models, preprocessing, threshold.
4	Frozen Cycle 25 evaluation	Evaluate unchanged pipeline on pre-defined Cycle-25 temporal blocks with same metrics.
5	Shift audit & controls	Measure feature/prevalence shifts; build bootstrap and temporal within-Cycle-24 nulls; run shift-aware sensitivity analyses.
6	SHAP drift analysis	Use one primary explanation protocol; compare magnitude, direction, ranks, overlap, and attribution distributions; correct multiple tests.
7	Interpretation & reproducibility	Apply pre-registered decision rules; package code, manifests, seeds, data provenance, models, and figures.

6. Predictive Evaluation Rules

- Primary metric: True Skill Statistic (TSS), always accompanied by active-region/entity-group bootstrap confidence intervals.
- Complementary metrics: HSS and/or F1, PR-AUC, Brier score and calibration diagnostics where probabilities are used.
- Classification threshold is selected using Cycle-24 validation only and then frozen.
- Cycle 25 is never used for hyperparameter tuning, preprocessing fitting, feature selection, threshold tuning, or model selection in the primary experiment.
- "Stable" versus "changed" predictive skill must be defined before examining Cycle-25 outcomes using a pre-registered equivalence/change rule (e.g. delta-TSS interval or equivalence margin justified from uncertainty/operational relevance).

7. Distribution-Shift and Null-Control Protocol

Evidence Ladder Before Declaring Explanation Drift

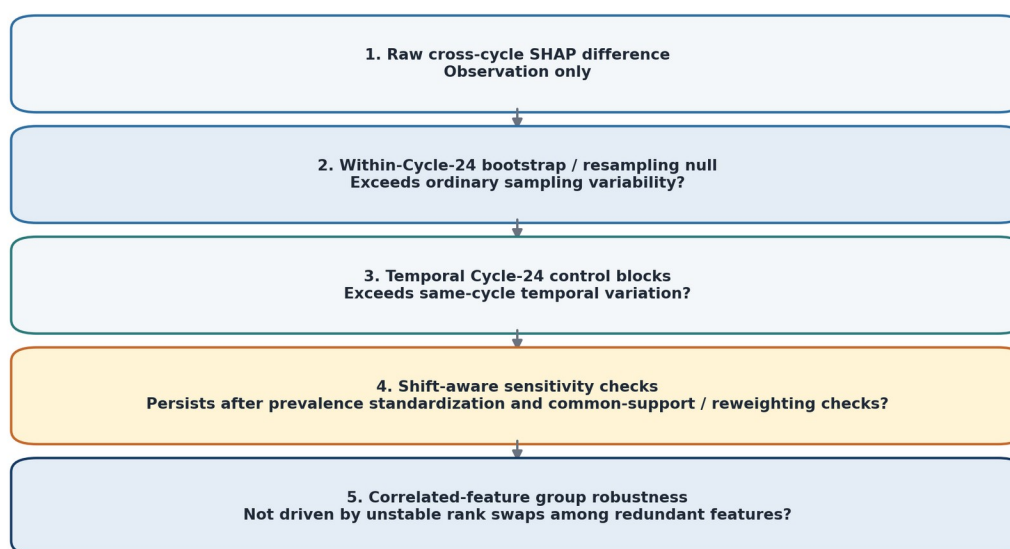


Figure 4. Evidence ladder required before declaring cross-cycle explanation drift.

- Audit marginal and selected joint feature distributions, missingness, disk position, region properties, and positive-label prevalence by cycle/block.
- Primary deployment-style comparison uses the observed Cycle-25 distribution.
- Class-conditional SHAP comparisons (positive and negative labels separately) reduce the risk that prevalence differences alone drive the result.
- Run prevalence-standardized sensitivity analysis and a common-support / domain-reweighting or matching sensitivity analysis where feasible.
- Build an entity-level bootstrap/resampling null within Cycle 24 using matched sample sizes.
- Add temporal same-cycle controls: compare separated Cycle-24 blocks/phases to quantify normal temporal explanation variation.

- Cross-cycle drift must exceed the relevant within-cycle null and remain directionally robust across pre-specified sensitivity analyses.

8. SHAP Protocol

Component	Rule
Model comparison unit	Same fixed model across Cycle 24 and Cycle 25. Cross-model ranking differences are secondary.
Explainers	Use model-appropriate SHAP explainer; exact library/version/settings recorded.
Reference/background	Primary reference sampled only from Cycle-24 training data; fixed across cycle evaluations; reasonable alternatives used only for sensitivity checks.
Global statistics	Mean SHAP , feature rank, top-k overlap, attribution distribution distance.
Directional statistics	SHAP sign/direction consistency and feature-response patterns where interpretable.
Inference	Empirical null from within-Cycle-24 controls; feature-wise tests corrected for multiple comparisons (e.g. FDR).
Correlations	Analyze individual features and pre-defined correlation groups; do not over-interpret rank swaps among redundant variables.

9. Pre-Registered Interpretation

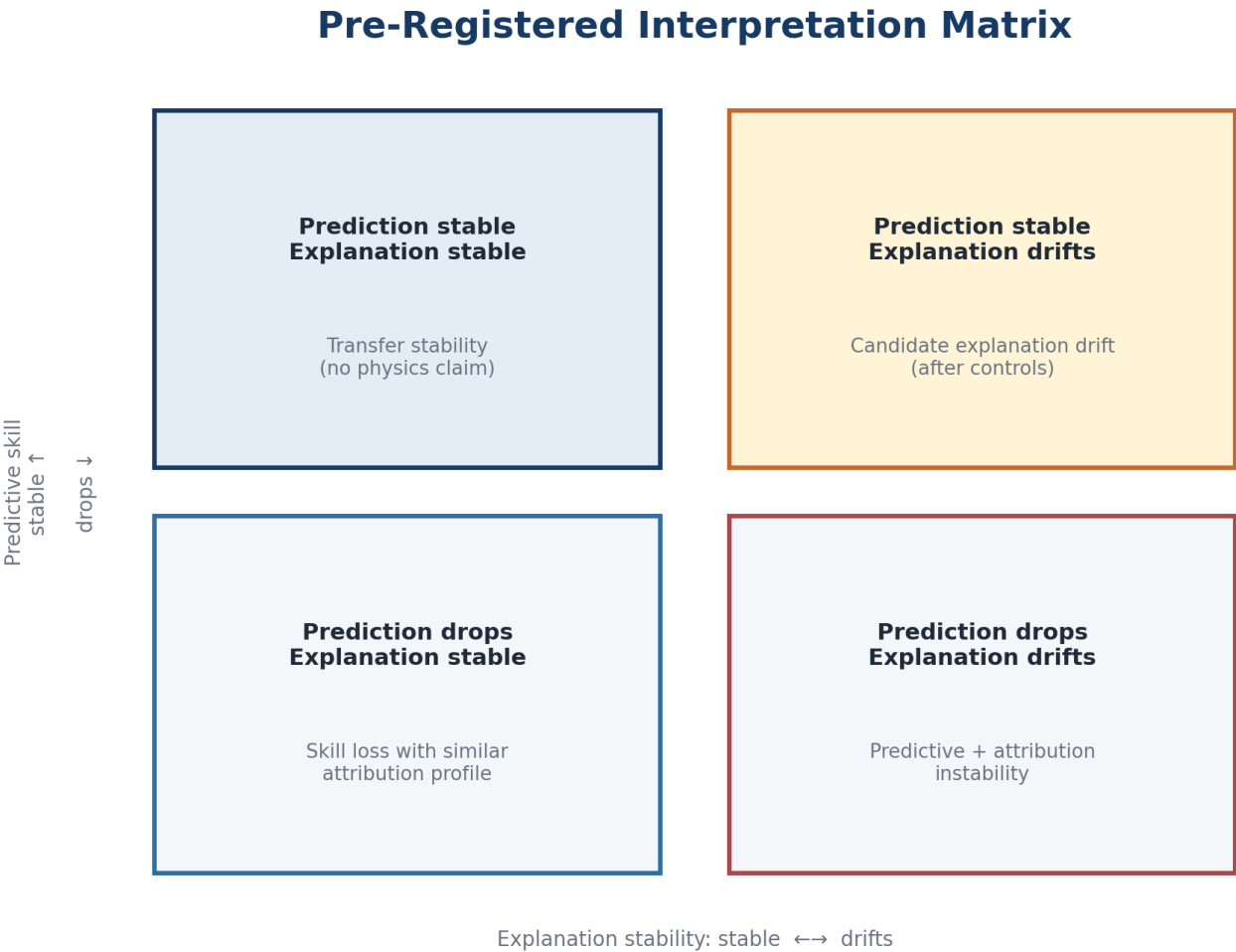


Figure 5. Interpretation matrix. Cells describe model evidence, not causal solar physics.

10. Claims Discipline

Do not claim	Use instead
First cross-cycle flare prediction	This study performs a specific frozen-model explanation-stability experiment across Cycles 24 and 25.
First SHAP flare paper	Prior SHAP flare forecasting exists; our contribution is cross-cycle attribution stability with controls.
Cycle 25 is necessarily OOD	Cycle 25 is a temporal/domain-shift evaluation; OOD language requires support diagnostics.
SHAP proves changed solar physics	SHAP describes feature attribution of the trained model on the evaluated inputs.
The frozen model changed internally	The model parameters are fixed; the observed attribution distribution may differ across input regimes.
Right for the wrong reasons	Possible attribution drift despite similar aggregate skill; causal interpretation requires stronger evidence.

11. Duck Checkpoint v2

Duck is a mandatory alignment audit against this v2 master. It must return ON TRACK, MISSING, RISKS / DEVIATIONS, NEXT ACTIONS, and a final decision: Continue / Fix before continuing / Re-scope.

Duck check	Question
Scope	Still testing predictive stability vs SHAP attribution stability for the same frozen model?
Data product	Definitive SHARP series, query/version metadata, quality rules, cadence/stride and label source locked?
Entity leakage	Connected HARP-NOAA entity groups isolated across partitions?
Cycle 25 isolation	No C25 use in fitting, tuning, threshold selection, feature selection or model choice?
Metrics	TSS + uncertainty + complementary metrics + calibration as appropriate?
Decision rules	Predictive "stable/change" and explanation-drift criteria pre-registered before viewing final C25 results?
Null controls	Both resampling and temporal within-Cycle-24 explanation baselines established?
Shift sensitivity	Prevalence/class-conditional and common-support/reweighting checks completed where applicable?
SHAP protocol	Same frozen model, same primary background/reference, versioned settings, sensitivity checks?
Correlated features	Feature-group robustness performed?
Claims	No causal physics or unsupported novelty language?
Reproducibility	Queries, manifests, seeds, environments, artifacts and figure scripts saved?

12. GitHub Repository Contract

The repository should preserve a reproducible research trail rather than only notebooks. Recommended top-level structure: docs/, configs/, data_manifests/, src/, experiments/, results/, figures/, tests/, environment files, and a CITATION.cff before public release.

Base-document rule

The three WHAT / WHY / HOW PDFs must cite this Master v2 and must not introduce a different target, data product, split unit, threshold policy, SHAP protocol, or decision rule without a versioned master update.

Reference Basis

These references support the current design. They are not the final literature review; novelty-sensitive claims must be re-checked before submission.

[1] Bobra et al. (2014), SHARP data products and HARP/NOAA mapping, Solar Physics. [Source](#)

[2] Hoeksema et al. (2014), HMI vector magnetic-field pipeline and definitive SHARP products, Solar Physics. [Source](#)

[3] Chen et al. (2019), SHARP flare precursors and correlated-feature instability, Space Weather. [Source](#)

[4] Goodwin et al. (2024), solar-cycle and training-window effects in simulated real-time flare forecasting, ApJ. [Source](#)

[5] Hu et al. (2025), data-source quality issues and Cycle-24-to-25 evaluation, arXiv:2512.13417. [Source](#)

[6] Wu et al. (2026), interpretable major-flare forecasting with SHAP, MNRAS. [Source](#)

[7] Li et al. (2026), explainable operational flare forecasting with SHAP, Space Weather. [Source](#)

[8] Roy et al. (2026), SuryaBench, Scientific Data. [Source](#)

[9] Liu et al. (2026), FlareDB for Solar Cycles 24 and 25, Scientific Data. [Source](#)